Complexity analysis for the oMDPi

1 Complexity

Theorem 1.1. *oMDPi is weakly NP-complete, for at least two downlink windows and even if priorities are fixed.*

Proof. oMDPi is in NP because the *Simulation Stop* algorithm can check a priority affectation and an interruption plan in polynomial time.

We give a construction π which is a polynomial reduction from an instance x of SubsetSum to an instance $\pi(x)$ of oMDPi. We assume (without loss of generality) that the instance x of SubsetSum is a set of real numbers $A = \{a_1, \ldots, a_n\}$ such that $\sum_{i=1}^{n} a_i = 1$, and hence is satisfiable if and only if there exists $S \subset A$ such that $\sum_{a_i \in S} = \frac{1}{2}$. The capacity of buffer $1 \leq i \leq n$ is $a_i(1+\frac{1}{m})$ with m a large number (at least so that $\min_i(a_i)/m < 1$). The capacity of buffer $n+1$ is another large number L (we can use $L = \max(2, n)$). All buffers are empty at the start of the first downlink. The first downlink window has dump rate $\delta = n$ and can be divided in 5 parts of length 1, starting at $t = 0$, followed by a part of duration L/n.

- In the first part, buffers $1 \leq i \leq n$ produce $1 + a_i + \frac{a_i}{m}$ and buffer $n+1$ produces $n - 1 - \frac{1}{m}$.
- In the second part, only buffer $n+1$ produces $L - \frac{1}{2}$.
- In the third part, the fill rate of buffer $n+1$ is null, and the fill rate of buffer $1 \leq i \leq n$ is $1 + a_i$.
- In the fourth part, the fill rate of buffer $n+1$ is n, and the fill rate of buffer $1 \leq i \leq n$ is null.
- In the fifth part, the fill rate of buffer $n+1$ is null, and the fill rate of buffer $1 \leq i \leq n$ is $\frac{a_i}{m}$.
- In the last part, the fill rate is null for all buffers.

The second downlink window has a dump rate $n+1$. It is divided in 5 parts the first, third and fifth of duration 1 and the second and fourth of duration $L/(n+1)$.

- In the first part, the fill rate of buffer $n+1$ is $L + n + 1 - \frac{1}{2} - \frac{1}{2m}$, and the fill rate of all other buffers is null.

- In the second part, the fill rate is null for all buffers.
- In the third part, the fill rate of buffer $n+1$ is $L+1$ and for $1 \leq i \leq n$, the fill rate of buffer i is $1 + a_i + \frac{a_i}{m}$.
- The fourth and fifth parts are the same as the second and third, respectively

We now prove that x is a satisfiable SubsetSum instance if and only if $\pi(x)$ is a satisfiable oMDPi instance. The following Lemmas are steps to prove Lemma 1.9 which is the "if" part of the proof. Lemma 1.10 is the "only if" part.

Lemma 1.2. *Buffers $1 \leq i \leq n$ must have equal priority, and buffer $n+1$ must have a strictly worse priority.*

Proof. Every buffer produces more than its capacity during the first downlink after the first part. Therefore, none of them can be interrupted before $t = 1$. Moreover, since the total production of buffers $1 \leq i \leq n$ on the first part is equal to the sum of the maximum that can be dumped (n) and of their total capacities ($a_i + \frac{a_i}{m}$), they must use all the bandwidth, hence buffer $n+1$ must be of strictly worse priority. Finally, since every buffer $1 \leq i \leq n$ must dump exactly 1, they must all have the same priority. □

Lemma 1.3. *At the start of the third part of the first downlink, buffer $n+1$ contains $L - \frac{1}{2}$ data, and all other buffers are empty.*

Proof. Simple application of the problem definition since we know the priority allocation by Lemma 1.2, and there is no interruption possible because every buffer has still too much data to produce at this point. □

Lemma 1.4. *Buffer $n+1$ must use all but $\frac{1}{2}$ of the bandwidth on the fourth part of the first downlink.*

Proof. By Lemma 1.2, other buffers use all the bandwidth on the third part and therefore its memory usage is still $U = L - \frac{1}{2}$ at the start of the fourth part (the same as the usage at the end of the second part by Lemma 1.3). Its production on this part is $P = n$ and its capacity is $C = L$. Let D the data dumped from buffer $n+1$ on the fourth part. We must have $U + P - D \leq C$, i.e., $D \geq n - \frac{1}{2}$. Since the dump rate is n and the part duration is 1, at most $\frac{1}{2}$ can be dumped by other buffers. □

Lemma 1.5. *A non-empty set $S \subseteq \{1, \ldots, n\}$ of buffers must be interrupted, and each interrupted buffer a_i cannot be interrupted earlier than $t = 3$ in the first downlink.*

Proof. Buffers $1 \leq i \leq n$ all have priority over buffer $n+1$ and have a total memory usage of 1. Therefore, if none of them is interrupted, they will transfer 1 over this part, which is a contradiction with Lemma 1.4.

Moreover, since the memory usage of buffer i is at most $a_i + \frac{a_i}{m}$, and since it dumped at rate 1, it cannot have been interrupted before time $t = 3$ since its usage at this point is a_i and $\frac{a_i}{m}$ will be produced later on. $\square$

By Lemma 1.5 there is a non-empty set of buffers I that have to be interrupted during the forth part. Let $0 < \alpha_i \leq a_i$ be the quantity of data in the buffer at time $t = 4$ and *not* dumped in the forth part.

Lemma 1.6. $\sum_{i \in S} \alpha_i \geq \frac{1}{2}$ *and* $\sum_{i \in S} a_i \geq \frac{1}{2}$

Proof. By Lemma 1.4, only $\frac{1}{2}$ data in total can be dumped by buffers $1 \leq i \leq n$ during part 4 of the first downlink. Since by Lemma 1.2 they all have the strictly better priority than buffer $n+1$ there would be a contradiction if $\sum_{i \in S} \alpha_i < \frac{1}{2}$. Since $\alpha_i \leq a_i$, we have $\sum_{i \in S} a_i \geq \frac{1}{2}$. $\square$

Lemma 1.7. *At time $t \geq 5$ in the first downlink*

- *the usage of buffer $i \in \{1, \ldots, n\} \setminus I$ is null;*
- *the usage of buffer $i \in S$ is $\alpha_i + \frac{a_i}{m}$;*

Proof. The total amount of data to dump from buffers $1 \leq i \leq n$ in the second part is at most $1 + \frac{1}{m} < n$ and since they have priority over the remaining buffer $n+1$ by Lemma 1.2, all the data held on an uninterrupted buffer will be dumped.

Now, buffer $i \in S$ has kept α_i of the data it had at time $t = 4$, and i(t is filled with an extra $\frac{a_1}{m}$ before time $t = 5$ (which it cannot dump since it is interrupted). $\square$

Lemma 1.8. *In the second downlink, all buffers have equal priority, and no interruption is possible before the fifth part.*

Proof. Each of the $n+1$ buffers produces its capacity plus one unit of data in the last part, and only $n+1$ units can be dumped. Hence they must all dump at least one unit and hence no interruption is possible before.

The same observation is true in the third part, and since no interruption is possible in this part, the only feasible priority allocation is to have all buffers at the same priority level.

Moreover, note that there is enough time in the second and fourth parts to empty all buffers. $\square$

Lemma 1.9. *If $\pi(x)$ is satisfiable, then x is satisfiable.*

Proof. In the second downlink window, buffers $i \in S$ begin with usage $\alpha_i + \frac{a_i}{m}$, and all other buffers are empty because of the long inactive sixth part of the first downlink. The overall initial usage of buffers $1 \leq i \leq n$ is less than $1 + \frac{1}{m}$. Since all buffers have the same priority by Lemma 1.8 a bandwidth of $n+1$ is sufficient for all buffers $1 \leq i \leq n$ to be empty at time $t = 1$ in the second downlink. Therefore, only buffer $n+1$ can have some non-null usage at that point. At the start of the downlink, the total usage is $U \geq \frac{1}{2} + \sum_{i \in S} \frac{a_i}{m}$. The sum of the production and of the initial usage minus the total data dump $n+1$ must be less than or equal to buffer $n+1$'s capacity, and therefore:

$$\frac{1}{2} + \sum_{i \in S} \frac{a_i}{m} + L + n + \frac{1}{2} - \frac{1}{2m} - (n+1) \leq L$$

that is,

$$\sum_{i \in S} a_i \leq \frac{1}{2}$$

By Lemma 1.6 we therefore have $\sum_{i \in S} a_i = \frac{1}{2}$, and hence I is a solution of the SubsetSum problem. □

Lemma 1.10. *If x is satisfiable, then $\pi(x)$ of is satisfiable.*

Proof. Only one priority assignment is allowed in our construction, so only the interruption decision matters. We simply interrupt buffer i at time $t = 3$ in the first downlink window and do no other interruption. Then we have a solution of oMDPi with $\alpha_i = a_i$ for all $1 \leq i \leq n$. □

The proof is complete as Lemma 1.9 and 1.10 show the equivalence of satisfiability of x and $\pi(x)$. □